

The Structure of Σ_t and Σ_t^{-1} in the AR(1) Diffusion Model

Explicit Inversion, Spectral Decomposition, and Numerical Study

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1 Motivation

In the previous note we derived the joint score of a diffusion model applied to an AR(1) sequence:

$$S(\mathbf{x}, t) = -\Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t), \quad (1)$$

where $\mathbf{x} = (x_0, \dots, x_K)^\top \in \mathbb{R}^{K+1}$ is the noisy trajectory, $\boldsymbol{\mu}_t = e^{-t}\boldsymbol{\mu}_a$ is the noisy mean, and

$$\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I_{K+1}, \quad \Delta_t = 1 - e^{-2t}. \quad (2)$$

The *entire* structure of the score is therefore encoded in the precision matrix Σ_t^{-1} . This document is devoted to a thorough analysis of Σ_t and Σ_t^{-1} : their explicit closed-form expressions, their spectral decomposition, and their numerical structure for a 20-element sequence at various diffusion times.

The overarching research question, as stated by Mézard during the meeting of March 18, 2026:

“What is the impact of the Markovian process underlying the sequence of frames on the structure of the score?”

In the clean limit ($t = 0$), the precision matrix Σ_0^{-1} is tridiagonal—the algebraic encoding of Markovianity. What happens as the diffusion noise t increases? Does the Markov structure survive, degrade gradually, or vanish abruptly? And in the eigenbasis of Σ_0 , can we make the entire problem fully independent?

2 Recap: The AR(1) Model and Its Matrices

We work with the scalar AR(1) Gaussian model from the previous note. The clean frames $\mathbf{a} = (a_0, \dots, a_K)^\top \in \mathbb{R}^{K+1}$ satisfy:

$$a_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad (3)$$

with $|\alpha| < 1$. The stationary variance is $\sigma_\infty^2 = \sigma_\eta^2/(1 - \alpha^2)$. Throughout this document, unless stated otherwise, we work in the **stationary regime** $\sigma_0^2 = \sigma_\infty^2$, which makes Σ_0 a symmetric Toeplitz matrix:

$$(\Sigma_0)_{jk} = \sigma_\infty^2 \alpha^{|j-k|}, \quad j, k = 0, \dots, K. \quad (4)$$

This simplification loses no generality for our structural analysis (the non-stationary case differs only in boundary corrections).

2.1 The clean precision matrix Σ_0^{-1}

The precision matrix is tridiagonal (Proposition 3.2 of the previous note):

$$\Sigma_0^{-1} = \frac{1}{\sigma_\eta^2} \begin{pmatrix} 1 + \alpha^2 & -\alpha & 0 & \cdots & 0 \\ -\alpha & 1 + \alpha^2 & -\alpha & \ddots & \vdots \\ 0 & -\alpha & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & 1 + \alpha^2 & -\alpha \\ 0 & \cdots & 0 & -\alpha & 1 \end{pmatrix}. \quad (5)$$

In the stationary case the $(0, 0)$ entry is $(1/\sigma_0^2 + \alpha^2/\sigma_\eta^2) = (1 + \alpha^2)/\sigma_\eta^2$, matching the interior entries. The only asymmetry is the (K, K) corner entry $1/\sigma_\eta^2$ instead of $(1 + \alpha^2)/\sigma_\eta^2$.

Remark 2.1 (Why is Σ_0^{-1} tridiagonal?). *For a jointly Gaussian vector, $(\Sigma_0^{-1})_{jk} = 0$ if and only if $a_j \perp a_k \mid \{a_\ell : \ell \neq j, k\}$. The AR(1) Markov property states that a_j and a_k are conditionally independent given the intermediate frames whenever $|j - k| \geq 2$. The tridiagonal structure of Σ_0^{-1} is therefore the exact algebraic encoding of Markovianity: the only non-zero entries are on the main diagonal and the first super/sub-diagonals, reflecting the fact that each frame is directly coupled only to its nearest neighbours.*

Remark 2.2 (Dense Σ_0 vs. sparse Σ_0^{-1}). *The covariance matrix Σ_0 is dense: every entry $\sigma_\infty^2 \alpha^{|j-k|}$ is non-zero for $\alpha \neq 0$. This is because $\text{Cov}(a_j, a_k)$ includes correlations mediated by all intermediate frames. The precision matrix strips away these indirect paths and retains only the direct couplings. This is a fundamental duality in Gaussian graphical models: the covariance encodes total (marginal) correlations; the precision encodes partial (conditional) correlations.*

3 Explicit Inversion: Closed Form for Σ_t^{-1}

3.1 The problem

Given $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$ from (2), we seek Σ_t^{-1} in a form that reveals the interplay between signal structure (Σ_0) and noise ($\Delta_t I$). We present four equivalent representations, each illuminating a different aspect.

3.2 Representation 1: Factored form via Σ_0^{-1}

Proposition 3.1 (Factored inversion formula). *Define the signal-to-noise ratio at diffusion time t :*

$$\gamma_t = \frac{e^{-2t}}{\Delta_t} = \frac{e^{-2t}}{1 - e^{-2t}}. \quad (6)$$

Then:

$$\Sigma_t^{-1} = \frac{1}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1} \quad (7)$$

Equivalently, factoring from the signal side:

$$\Sigma_t^{-1} = e^{2t} (\Sigma_0 + (e^{2t} - 1) I)^{-1} \quad (8)$$

Proof. Factor Δ_t from Σ_t :

$$\Sigma_t = \Delta_t \left(\frac{e^{-2t}}{\Delta_t} \Sigma_0 + I \right) = \Delta_t (I + \gamma_t \Sigma_0).$$

Inverting: $\Sigma_t^{-1} = \frac{1}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1}$.

For (8), factor e^{-2t} instead:

$$\Sigma_t = e^{-2t} \left(\Sigma_0 + \frac{\Delta_t}{e^{-2t}} I \right).$$

Using $\Delta_t/e^{-2t} = (1 - e^{-2t})/e^{-2t} = e^{2t} - 1$, we get $\Sigma_t = e^{-2t}(\Sigma_0 + (e^{2t} - 1)I)$ and the result follows. $\square$

Remark 3.2 (Physical interpretation of γ_t). *The ratio $\gamma_t = e^{-2t}/\Delta_t$ is the ratio of the signal coefficient e^{-2t} to the noise coefficient Δ_t in the decomposition $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$. It satisfies:*

- $\gamma_t \rightarrow +\infty$ as $t \rightarrow 0^+$ (signal dominates),
- $\gamma_t = 1$ when $e^{-2t} = 1/2$, i.e. $t = \frac{1}{2} \ln 2 \approx 0.347$,
- $\gamma_t \rightarrow 0$ as $t \rightarrow +\infty$ (noise dominates).

When $\gamma_t \gg 1$, the term $\gamma_t \Sigma_0$ dominates inside $(I + \gamma_t \Sigma_0)^{-1}$, and the inverse is approximately $(\gamma_t \Sigma_0)^{-1} = \Sigma_0^{-1}/\gamma_t$, yielding $\Sigma_t^{-1} \approx \Sigma_0^{-1}/(e^{-2t})$ —the tridiagonal structure is preserved. When $\gamma_t \ll 1$, the I dominates and $\Sigma_t^{-1} \approx I/\Delta_t$ —a diagonal matrix, completely decoupled.

3.3 Representation 2: In terms of Σ_0^{-1} (Mézard’s form)

During the meeting, Mézard wrote the inverse as:

$$\Sigma_t^{-1} = \frac{\Sigma_0^{-1}}{e^{-2t} I + \Delta_t \Sigma_0^{-1}}, \quad (9)$$

where the “division” means $\Sigma_0^{-1} \cdot (e^{-2t} I + \Delta_t \Sigma_0^{-1})^{-1}$.

Proposition 3.3 (Mézard’s matrix form).

$$\boxed{\Sigma_t^{-1} = \Sigma_0^{-1} (e^{-2t} I + \Delta_t \Sigma_0^{-1})^{-1} = (e^{-2t} I + \Delta_t \Sigma_0^{-1})^{-1} \Sigma_0^{-1}} \quad (10)$$

The two orderings are equal because Σ_0^{-1} commutes with $(e^{-2t} I + \Delta_t \Sigma_0^{-1})$.

Proof. Left-multiply $\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I$ by Σ_0^{-1} :

$$\Sigma_0^{-1} \Sigma_t = e^{-2t} I + \Delta_t \Sigma_0^{-1}.$$

Therefore $\Sigma_t^{-1} = (\Sigma_0^{-1} \Sigma_t)^{-1} \Sigma_0^{-1} = (e^{-2t} I + \Delta_t \Sigma_0^{-1})^{-1} \Sigma_0^{-1}$.

For commutativity: let $B = e^{-2t} I + \Delta_t \Sigma_0^{-1}$. Then B is a polynomial in Σ_0^{-1} (of degree 1), so B commutes with Σ_0^{-1} , and hence B^{-1} commutes with Σ_0^{-1} . $\square$

Remark 3.4 (Why this fills in the tridiagonal zeros). *Formula (10) writes Σ_t^{-1} as a product of Σ_0^{-1} (tridiagonal) with $(e^{-2t} I + \Delta_t \Sigma_0^{-1})^{-1}$. The matrix $e^{-2t} I + \Delta_t \Sigma_0^{-1}$ is itself tridiagonal (a diagonal perturbation of a tridiagonal matrix). However, the inverse of a tridiagonal matrix is generically dense: if T is an $n \times n$ irreducible tridiagonal matrix, then $(T^{-1})_{jk} \neq 0$ for all j, k .*

This is the mechanism by which diffusion noise destroys the Markov sparsity. The “denominator” $(e^{-2t} I + \Delta_t \Sigma_0^{-1})^{-1}$ is a full matrix. Multiplying the tridiagonal Σ_0^{-1} by this full matrix produces a full matrix Σ_t^{-1} .

When Δ_t is small, the denominator is approximately $(e^{-2t} I)^{-1} = e^{2t} I$, and the product $\Sigma_0^{-1} \cdot e^{2t} I = e^{2t} \Sigma_0^{-1}$ is still tridiagonal. This is why the Markov structure is preserved at short diffusion times and degraded at long diffusion times.

3.4 Representation 3: Woodbury expansion

For analytical work, the Woodbury matrix identity provides a series-like expansion.

Proposition 3.5 (Woodbury form).

$$\Sigma_t^{-1} = \frac{1}{\Delta_t} I - \frac{e^{-2t}}{\Delta_t^2} \Sigma_0 (\Sigma_0^{-1} + \frac{e^{-2t}}{\Delta_t} I)^{-1}. \quad (11)$$

Proof. Apply the Woodbury identity $(A + UCU^\top)^{-1} = A^{-1} - A^{-1}U(C^{-1} + U^\top A^{-1}U)^{-1}U^\top A^{-1}$ with $A = \Delta_t I$, $U = I$, $C = e^{-2t} \Sigma_0$:

$$\Sigma_t^{-1} = \frac{1}{\Delta_t} I - \frac{1}{\Delta_t} I \cdot \left(\frac{1}{e^{-2t}} \Sigma_0^{-1} + \frac{1}{\Delta_t} I \right)^{-1} \cdot \frac{1}{\Delta_t} I.$$

Simplifying the inner inverse and collecting factors gives (11). $\square$

Remark 3.6 (Perturbative reading). *Rewrite (11) as*

$$\Sigma_t^{-1} = \frac{1}{\Delta_t} I - \frac{\gamma_t}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1} \Sigma_0.$$

At large t ($\gamma_t \rightarrow 0$), the second term vanishes and $\Sigma_t^{-1} \rightarrow (1/\Delta_t)I \rightarrow I$. At small t ($\gamma_t \rightarrow \infty$), the second term dominates and one recovers the clean precision. The Woodbury form thus naturally organises the “noise floor” (I/Δ_t) and the “signal correction” (the second term).

3.5 Representation 4: Spectral form (the definitive formula)

Theorem 3.7 (Spectral decomposition of Σ_t^{-1}). *Let $\Sigma_0 = U D U^\top$ be the eigendecomposition of Σ_0 , where U is orthogonal and $D = \text{diag}(d_0, d_1, \dots, d_K)$ with $d_0 \geq d_1 \geq \dots \geq d_K > 0$. Then for all $t \geq 0$:*

$$\Sigma_t^{-1} = U \text{diag}\left(\frac{1}{e^{-2t}d_0 + \Delta_t}, \frac{1}{e^{-2t}d_1 + \Delta_t}, \dots, \frac{1}{e^{-2t}d_K + \Delta_t}\right) U^\top \quad (12)$$

Equivalently, writing $\mathbf{u}_m$ for the m -th column of U (the m -th eigenvector of Σ_0):

$$\Sigma_t^{-1} = \sum_{m=0}^K \frac{1}{e^{-2t}d_m + \Delta_t} \mathbf{u}_m \mathbf{u}_m^\top. \quad (13)$$

Proof. Since $\Sigma_0 = U D U^\top$:

$$\Sigma_t = e^{-2t} U D U^\top + \Delta_t U U^\top = U (e^{-2t} D + \Delta_t I) U^\top.$$

The matrix $e^{-2t} D + \Delta_t I$ is diagonal with entries $e^{-2t} d_m + \Delta_t$. Inverting:

$$\Sigma_t^{-1} = U (e^{-2t} D + \Delta_t I)^{-1} U^\top = U \text{diag}\left(\frac{1}{e^{-2t}d_m + \Delta_t}\right)_{m=0}^K U^\top. \quad \square$$

Critical observation: shared eigenvectors

The eigenvectors of Σ_t and Σ_t^{-1} are those of Σ_0 , for *all* diffusion times t . Only the eigenvalues change with t .

Why? Because $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$ is a linear combination of Σ_0 and I . Adding a multiple of the identity to a matrix shifts all eigenvalues by the same amount but leaves every eigenvector unchanged. Rescaling by e^{-2t} scales all eigenvalues but again preserves eigenvectors.

This is a direct consequence of the *Gaussianity* of $P_0(\mathbf{a})$, not of Markovianity. For any Gaussian P_0 (Markov or not), the noise $\Delta_t I$ always combines with Σ_0 via addition, preserving the eigenbasis.

4 Eigenvalues of Σ_t and Σ_t^{-1} : Explicit Formulas

4.1 Eigenvalues of Σ_0 for the stationary AR(1) chain

In the stationary case, Σ_0 is a symmetric Toeplitz matrix with entries $\sigma_\infty^2 \alpha^{|j-k|}$. Its precision Σ_0^{-1} is tridiagonal with constant interior diagonal $(1 + \alpha^2)/\sigma_\eta^2$ and constant off-diagonal $-\alpha/\sigma_\eta^2$, plus a boundary correction at (K, K) .

For a truly periodic (circulant) version of this matrix, the eigenvalues and eigenvectors are exact Fourier modes. For the finite chain with free boundary conditions, they are given by discrete sine-like functions.

Proposition 4.1 (Eigenvalues of the stationary precision matrix, interior approximation). *For $K \gg 1$, the eigenvalues of the interior part of Σ_0^{-1} (i.e. ignoring the boundary correction at position (K, K)) are:*

$$\lambda_m^{(\text{prec})} = \frac{1}{\sigma_\eta^2} (1 + \alpha^2 - 2\alpha \cos \theta_m), \quad \theta_m = \frac{m\pi}{K+1}, \quad m = 1, \dots, K+1, \quad (14)$$

and the corresponding eigenvalues of Σ_0 are:

$$d_m = \frac{\sigma_\eta^2}{1 + \alpha^2 - 2\alpha \cos \theta_m} = \frac{\sigma_\eta^2}{(1 - \alpha)^2 + 2\alpha(1 - \cos \theta_m)}. \quad (15)$$

The eigenvectors are approximately $u_m(k) \propto \sin(k \theta_m + \phi_m)$ for suitable phases ϕ_m .

Remark 4.2 (Interpretation of the eigenvalue spectrum). *The eigenvalue (15) is a “spectral density” in the Fourier variable θ .*

- **Largest eigenvalue** ($m = 1, \theta_1 \approx 0$): $d_1 \approx \sigma_\eta^2/(1 - \alpha)^2 = \sigma_\infty^2/(1 - \alpha)$. For α close to 1, this is very large—it corresponds to the “slow mode,” a nearly constant eigenvector where all frames move together. This mode captures the long-range correlations of the AR(1) chain.
- **Smallest eigenvalue** ($m = K + 1, \theta_{K+1} \approx \pi$): $d_{K+1} \approx \sigma_\eta^2/(1 + \alpha)^2$. This is the “fast mode,” an alternating-sign eigenvector. It captures the highest-frequency fluctuations.
- **Spectral ratio**: $d_{\max}/d_{\min} \approx (1 + \alpha)^2/(1 - \alpha)^2$. For $\alpha = 0.9$, this ratio is $(1.9/0.1)^2 = 361$, indicating a highly anisotropic covariance.

4.2 Eigenvalues of Σ_t and Σ_t^{-1} as functions of t

Corollary 4.3 (Eigenvalue evolution). *The m -th eigenvalue of Σ_t and Σ_t^{-1} are:*

$$\lambda_m(t) = e^{-2t} d_m + \Delta_t, \quad \lambda_m^{-1}(t) = \frac{1}{e^{-2t} d_m + \Delta_t}. \quad (16)$$

Remark 4.4 (All eigenvalues converge to 1). *As $t \rightarrow \infty$, $e^{-2t} \rightarrow 0$ and $\Delta_t \rightarrow 1$, so $\lambda_m(t) \rightarrow 1$ for every m . The covariance $\Sigma_t \rightarrow I$ (isotropic) and the precision $\Sigma_t^{-1} \rightarrow I$ (decoupled). The rate of convergence depends on d_m : modes with large d_m (slow modes) retain signal longer, while modes with small d_m (fast modes) are overwhelmed by noise earlier.*

Quantitatively, the “half-life” of the m -th mode—the time at which half the eigenvalue comes from signal and half from noise—satisfies:

$$e^{-2t_m^*} d_m = \Delta_{t_m^*}, \quad \implies \quad t_m^* = \frac{1}{2} \ln(1 + d_m). \quad (17)$$

For the slow mode ($d_0 \gg 1$), $t_0^ \approx \frac{1}{2} \ln d_0 \gg 1$. For the fast mode ($d_K \ll 1$ when $\alpha \approx 1$), $t_K^* \approx d_K/2 \ll 1$.*

5 The Score in the Eigenbasis: Full Independence

This section develops the key insight from the meeting: in the eigenbasis of Σ_0 , the reverse diffusion process decouples into $K + 1$ independent scalar problems.

5.1 Change of basis

Define the rotated variables:

$$\tilde{\mathbf{x}} = U^\top \mathbf{x}, \quad \tilde{\boldsymbol{\mu}}_t = U^\top \boldsymbol{\mu}_t, \quad \tilde{\mathbf{a}} = U^\top \mathbf{a}, \quad (18)$$

where U is the matrix of eigenvectors of Σ_0 (as in Theorem 3.7).

Theorem 5.1 (Decoupling in the eigenbasis). *In the rotated basis $\tilde{\mathbf{x}} = U^\top \mathbf{x}$:*

(i) *The joint score becomes component-wise independent:*

$$\tilde{S}_m(\tilde{\mathbf{x}}, t) = -\frac{\tilde{x}_m - \tilde{\mu}_{t,m}}{e^{-2t} d_m + \Delta_t}, \quad m = 0, 1, \dots, K. \quad (19)$$

Each component $\tilde{S}_m$ depends only on $\tilde{x}_m$, not on any other component.

(ii) *The noisy distribution factorises:*

$$P_t(\tilde{\mathbf{x}}) = \prod_{m=0}^K \frac{1}{\sqrt{2\pi\lambda_m(t)}} \exp\left(-\frac{(\tilde{x}_m - \tilde{\mu}_{t,m})^2}{2\lambda_m(t)}\right), \quad (20)$$

where $\lambda_m(t) = e^{-2t} d_m + \Delta_t$. The modes are independent Gaussians.

(iii) *The reverse SDE decouples:*

$$d\tilde{x}_m = \left[\tilde{x}_m + 2\tilde{S}_m(\tilde{x}_m, t) \right] dt + \sqrt{2} d\tilde{W}_m, \quad (21)$$

where $\tilde{W}_m = (U^\top \mathbf{W})_m$ are independent Brownian motions (since U is orthogonal and the original $\mathbf{W}$ has independent components).

Proof. The score in the original basis is $S(\mathbf{x}, t) = -\Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t)$. Under $\tilde{\mathbf{x}} = U^\top \mathbf{x}$:

$$\tilde{S}(\tilde{\mathbf{x}}, t) = U^\top S(U\tilde{\mathbf{x}}, t) = -U^\top \Sigma_t^{-1} U (\tilde{\mathbf{x}} - \tilde{\boldsymbol{\mu}}_t).$$

By Theorem 3.7, $U^\top \Sigma_t^{-1} U = \text{diag}(1/\lambda_m(t))$, giving (19).

For (ii), the Gaussian $P_t(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_t, \Sigma_t)$ under the orthogonal change of variables $\tilde{\mathbf{x}} = U^\top \mathbf{x}$ becomes $P_t(\tilde{\mathbf{x}}) = \mathcal{N}(\tilde{\mathbf{x}}; U^\top \boldsymbol{\mu}_t, U^\top \Sigma_t U) = \mathcal{N}(\tilde{\mathbf{x}}; \tilde{\boldsymbol{\mu}}_t, \Lambda(t))$ where $\Lambda(t) = \text{diag}(\lambda_m(t))$ is diagonal, hence the components are independent.

For (iii), orthogonal rotation of i.i.d. standard Brownian motions $\mathbf{W}$ yields i.i.d. standard Brownian motions $\tilde{\mathbf{W}} = U^\top \mathbf{W}$ (isometric invariance of the Wiener process). $\square$

The eigenbasis makes the Gaussian problem trivial

In the eigenbasis of Σ_0 , the $K+1$ -dimensional diffusion problem splits into $K+1$ completely independent scalar Ornstein–Uhlenbeck processes. The m -th mode is a 1D Gaussian with variance $\lambda_m(t) = e^{-2t}d_m + \Delta_t$ and score $\tilde{S}_m = -(\tilde{x}_m - \tilde{\mu}_{t,m})/\lambda_m(t)$.

Crucially, this decoupling is a consequence of Gaussianity, not of Markovianity. Any Gaussian P_0 (Markov or not) admits the same diagonalisation. The Markov structure affects *which* eigenvectors U arise (spatially localised, sine-like modes for the tridiagonal precision), and *how* the eigenvalues d_m are distributed (the specific spectral formula (15)), but the existence of a decoupling basis is a generic Gaussian property.

For non-Gaussian P_0 , this decoupling breaks down: the score becomes a nonlinear function of $\tilde{\mathbf{x}}$, and the components couple through higher-order statistics.

6 Numerical Study: $K = 19$ (20 Frames)

We now examine Σ_t and Σ_t^{-1} numerically for a 20-frame sequence with parameters:

$$K = 19, \quad \alpha = 0.9, \quad \sigma_0^2 = 1, \quad \sigma_\eta^2 = 1 - \alpha^2 = 0.19 \quad (\text{stationary: } \sigma_\infty^2 = 1). \quad (22)$$

6.1 Σ_0 and Σ_0^{-1} : dense vs. tridiagonal

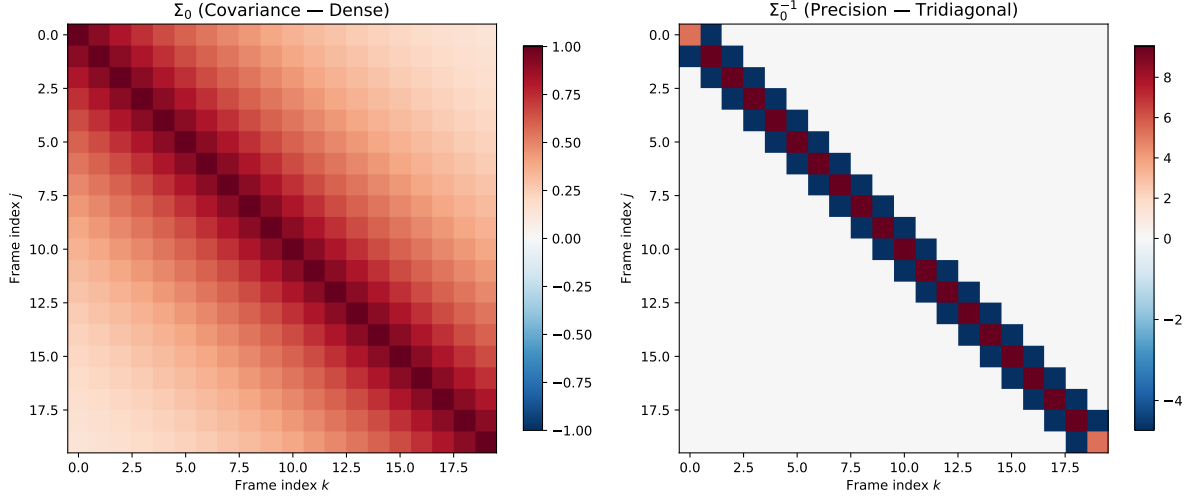


Figure 1: Left: the covariance Σ_0 is dense, with entries decaying as $\alpha^{|j-k|}$. Right: the precision Σ_0^{-1} is tridiagonal, encoding only nearest-neighbour couplings. The contrast illustrates the duality between marginal correlations (covariance) and conditional correlations (precision).

6.2 Explicit 5×5 matrices at selected diffusion times

To build intuition, we display the full numerical values for $K = 4$ (5 frames) at four diffusion times. All entries are rounded to 4 decimal places.

At $t = 0$ ($\Delta_t = 0$, pure signal):

$$\Sigma_0^{-1} = \begin{pmatrix} 5.2632 & -4.7368 & 0 & 0 & 0 \\ -4.7368 & 9.5263 & -4.7368 & 0 & 0 \\ 0 & -4.7368 & 9.5263 & -4.7368 & 0 \\ 0 & 0 & -4.7368 & 9.5263 & -4.7368 \\ 0 & 0 & 0 & -4.7368 & 5.2632 \end{pmatrix}. \quad (23)$$

Perfectly tridiagonal. The zero entries encode conditional independence: $a_j \perp a_k \mid \text{rest}$ for $|j-k| \geq 2$.

At $t = 0.3$ ($\Delta_t = 0.4512$):

$$\Sigma_t^{-1} = \begin{pmatrix} 1.4691 & -0.4530 & -0.2801 & -0.1823 & -0.1332 \\ -0.4530 & 1.5967 & -0.3831 & -0.2493 & -0.1823 \\ -0.2801 & -0.3831 & 1.6275 & -0.3831 & -0.2801 \\ -0.1823 & -0.2493 & -0.3831 & 1.5967 & -0.4530 \\ -0.1332 & -0.1823 & -0.2801 & -0.4530 & 1.4691 \end{pmatrix}. \quad (24)$$

The tridiagonal zeros are now filled, but with entries significantly smaller than the tridiagonal band. The first off-diagonal (≈ -0.45) is still $\sim 2\times$ larger than the second off-diagonal (≈ -0.25).

At $t = 1.0$ ($\Delta_t = 0.8647$):

$$\Sigma_t^{-1} = \begin{pmatrix} 1.0345 & -0.1014 & -0.0852 & -0.0728 & -0.0636 \\ -0.1014 & 1.0405 & -0.0975 & -0.0833 & -0.0728 \\ -0.0852 & -0.0975 & 1.0424 & -0.0975 & -0.0852 \\ -0.0728 & -0.0833 & -0.0975 & 1.0405 & -0.1014 \\ -0.0636 & -0.0728 & -0.0852 & -0.1014 & 1.0345 \end{pmatrix}. \quad (25)$$

Near-diagonal. All off-diagonal entries are of similar magnitude—the tridiagonal prominence is lost.

At $t = 3.0$ ($\Delta_t = 0.9975$):

$$\Sigma_t^{-1} \approx \begin{pmatrix} 1.0000 & -0.0022 & -0.0020 & -0.0018 & -0.0016 \\ -0.0022 & 1.0000 & -0.0022 & -0.0020 & -0.0018 \\ -0.0020 & -0.0022 & 1.0000 & -0.0022 & -0.0020 \\ -0.0018 & -0.0020 & -0.0022 & 1.0000 & -0.0022 \\ -0.0016 & -0.0018 & -0.0020 & -0.0022 & 1.0000 \end{pmatrix}. \quad (26)$$

Essentially the identity. All inter-frame coupling is below 0.3%. The score is $S(\mathbf{x}, t) \approx -\mathbf{x}/\Delta_t \approx -\mathbf{x}$, each component independent.

6.3 Σ_t and Σ_t^{-1} heatmaps for $K = 19$

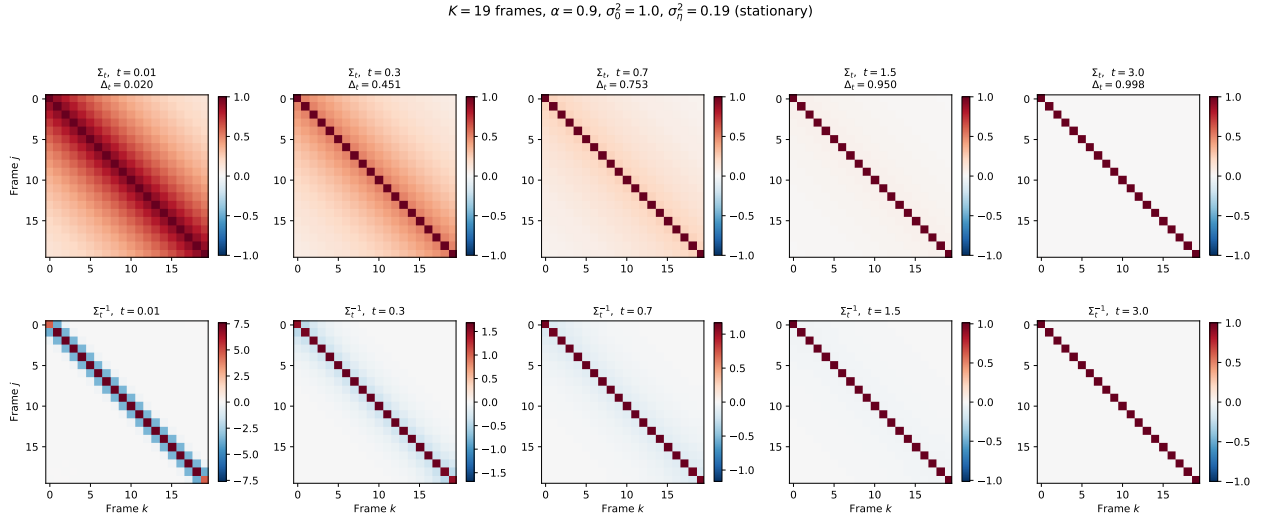


Figure 2: Top row: Σ_t at $t = 0.01, 0.3, 0.7, 1.5, 3.0$. The off-diagonal correlations shrink as e^{-2t} . Bottom row: Σ_t^{-1} at the same times. At $t = 0.01$, the precision is visually tridiagonal. By $t = 0.3$, the structural zeros are filled but the tridiagonal band remains dominant. By $t = 3.0$, the matrix is essentially I .

6.4 How much of Σ_t^{-1} lives on the tridiagonal band?

To quantify the persistence of Markov structure, define the *tridiagonal weight fraction*:

$$\rho_{\text{tri}}(t) = \frac{\sum_{|j-k| \leq 1} |(\Sigma_t^{-1})_{jk}|}{\sum_{j,k} |(\Sigma_t^{-1})_{jk}|}. \quad (27)$$

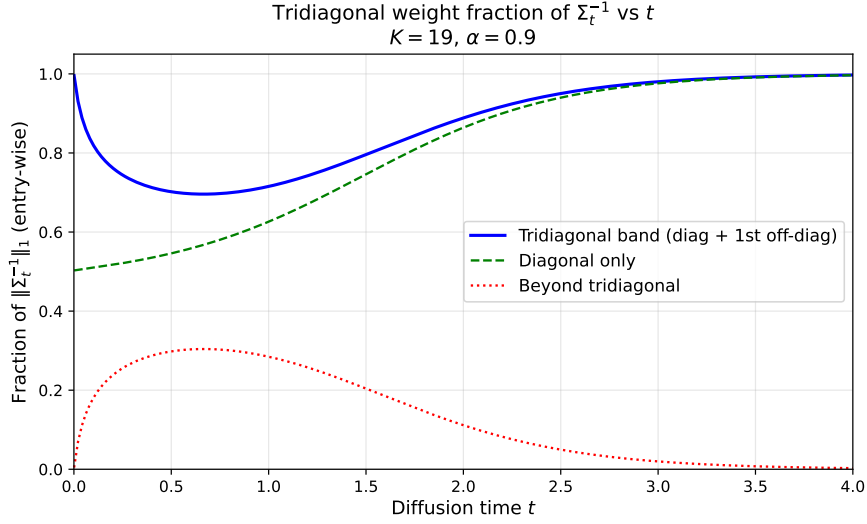


Figure 3: Tridiagonal weight fraction $\rho_{\text{tri}}(t)$ as a function of diffusion time. At $t = 0$: $\rho_{\text{tri}} = 1$ (perfectly tridiagonal). The fraction decays monotonically, but the tridiagonal band retains $> 80\%$ of the total weight until $t \approx 0.3$. At large t , the diagonal alone captures nearly all the weight (green dashed), as the matrix approaches I .

6.5 Row profile: coupling strength as a function of distance

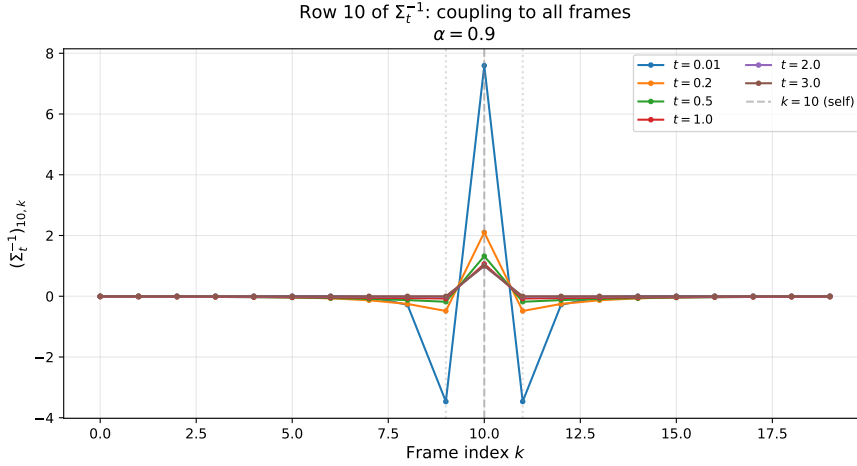


Figure 4: Row 10 of Σ_t^{-1} (the middle frame's coupling to all other frames) at several diffusion times. At $t = 0.01$, the coupling is concentrated on the nearest neighbours. At $t = 0.2$, the coupling has spread to further frames, decaying approximately exponentially with distance. At large t , all couplings vanish.

6.6 Off-diagonal decay: exponential envelope

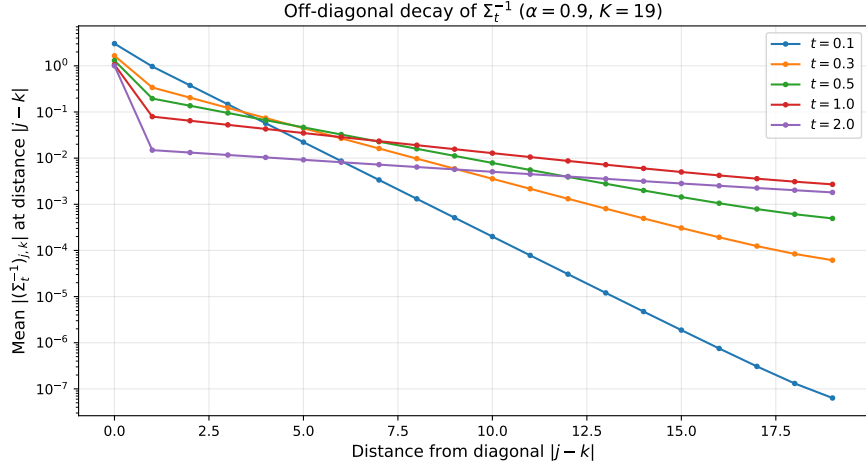


Figure 5: Mean absolute value of $(\Sigma_t^{-1})_{jk}$ as a function of the distance $|j - k|$ from the diagonal (log scale). At all positive t , the off-diagonal entries decay approximately exponentially with distance, but the decay rate depends on t : slower decay at small t (more long-range coupling).

7 Eigenvalue Analysis

7.1 Spectrum of Σ_0

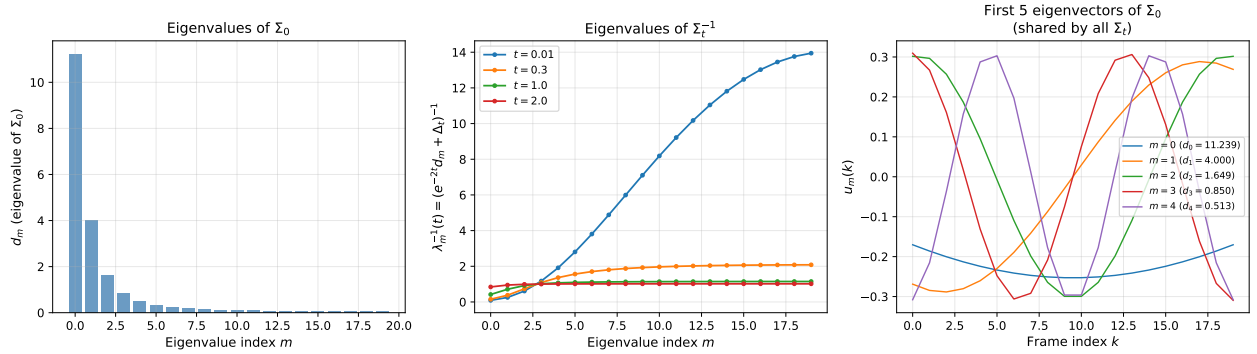


Figure 6: Left: eigenvalues of Σ_0 , ordered from largest to smallest. The slow mode ($m = 0$) has $d_0 \approx 10$, while the fast mode ($m = 19$) has $d_{19} \approx 0.028$. Middle: eigenvalues of Σ_t^{-1} at several t values; all converge to 1 as $t \rightarrow \infty$. Right: the first 5 eigenvectors of Σ_0 —smooth, sine-like oscillations, with the m -th eigenvector having approximately m zero crossings.

7.2 Eigenvalue trajectories vs. diffusion time

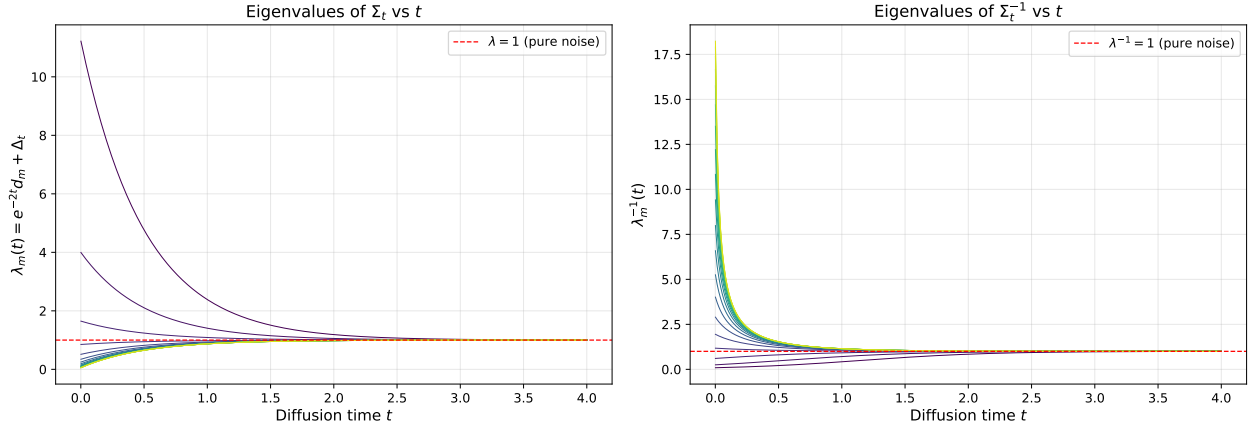


Figure 7: Left: eigenvalues of Σ_t as functions of t . Each curve $\lambda_m(t) = e^{-2t}d_m + \Delta_t$ interpolates from d_m (at $t = 0$) to 1 (as $t \rightarrow \infty$). The slow modes (large d_m) approach 1 from above; the fast modes (small d_m) approach 1 from below. Right: the reciprocals $\lambda_m^{-1}(t)$, which are the eigenvalues of Σ_t^{-1} . These control the score: large $\lambda_m^{-1}(t)$ means the m -th mode contributes strongly to the score.

7.3 Σ_t^{-1} in the eigenbasis: perfect diagonality

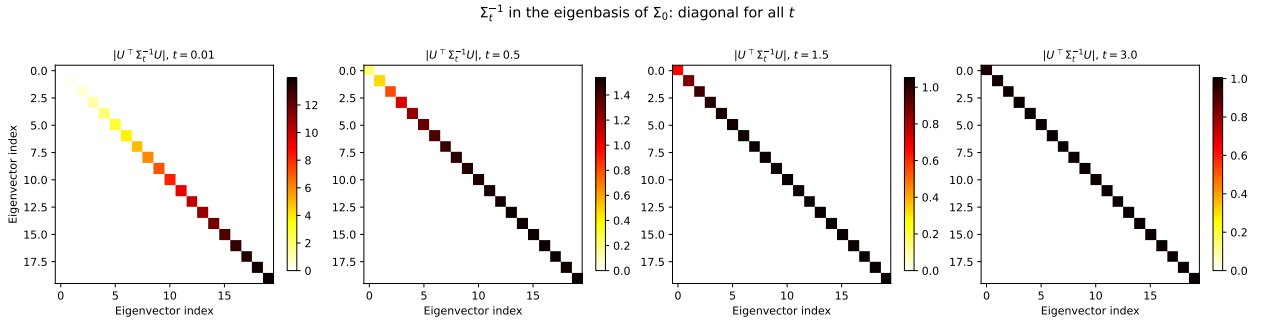


Figure 8: The matrix $|U^T \Sigma_t^{-1} U|$ at four diffusion times. It is *exactly* diagonal at all t , confirming that the eigenvectors of Σ_0 diagonalise Σ_t^{-1} for all t simultaneously. The diagonal entries are $1/(e^{-2t}d_m + \Delta_t)$.

7.4 Condition number

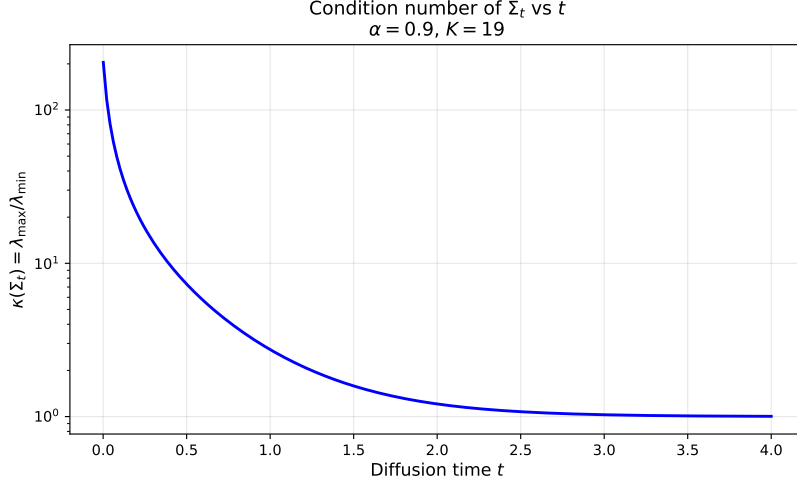


Figure 9: Condition number $\kappa(\Sigma_t) = \lambda_{\max}(t)/\lambda_{\min}(t)$ as a function of t . At $t = 0$, $\kappa = d_0/d_K \approx 361$ (highly anisotropic). As $t \rightarrow \infty$, $\kappa \rightarrow 1$ (isotropic). The condition number quantifies how “stretched” the Gaussian ellipsoid is: large κ means the score is very sensitive to perturbations along the fast-mode directions.

8 Sensitivity to α

The AR(1) parameter α controls the strength of inter-frame correlation. We examine how Σ_t and Σ_t^{-1} change as α varies.

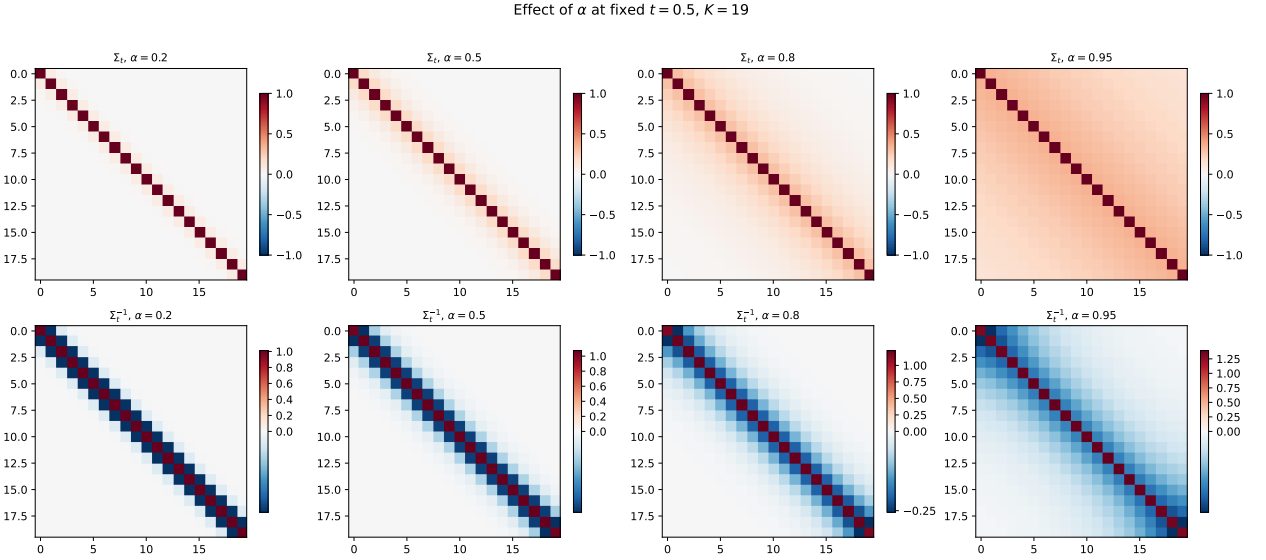


Figure 10: Σ_t (top) and Σ_t^{-1} (bottom) at fixed $t = 0.5$ for $\alpha \in \{0.2, 0.5, 0.8, 0.95\}$. For small α , the clean covariance is already nearly diagonal, so diffusion noise has little structural effect. For large α , the clean covariance is highly correlated, and the precision at $t = 0.5$ retains significant off-tridiagonal weight—the Markov structure is more persistent in the score for strongly correlated sequences.

Observation 8.1 (α controls the “lifetime” of Markov structure in the score). *For large α , the eigenvalue spread $d_{\max}/d_{\min} \sim (1 + \alpha)^2/(1 - \alpha)^2$ is large. The slow mode has half-life $t_0^* \approx \frac{1}{2} \ln d_0$, which grows as $\alpha \rightarrow 1$. Therefore, for strongly correlated AR(1) chains, the score retains inter-frame coupling to larger diffusion times.*

Conversely, for small α (weakly correlated), the eigenvalues cluster near $\sigma_\infty^2 \approx 1$, the condition number is small, and the Markov structure in the score is washed out quickly.

9 Component-Wise Score Structure

Combining the algebraic and numerical results, we can now give a complete picture of the k -th component of the joint score.

9.1 In the original frame basis

From Corollary 5.1 of the previous note:

$$S_k(\mathbf{x}, t) = - \sum_{j=0}^K (\Sigma_t^{-1})_{kj} (x_j - \mu_{t,j}). \quad (28)$$

At $t = 0$, $\Sigma_t^{-1} = \Sigma_0^{-1}$ is tridiagonal, so:

$$S_k(\mathbf{x}, 0) = -(\Sigma_0^{-1})_{k,k-1}(x_{k-1} - \mu_{a,k-1}) - (\Sigma_0^{-1})_{k,k}(x_k - \mu_{a,k}) - (\Sigma_0^{-1})_{k,k+1}(x_{k+1} - \mu_{a,k+1}). \quad (29)$$

This is a **nearest-neighbour coupling**: the score at frame k depends only on frames $k-1, k, k+1$. This is the Markov structure manifesting in the score.

For $t > 0$, Σ_t^{-1} is dense, and S_k couples to *all* frames. However, the coupling strength $|(\Sigma_t^{-1})_{kj}|$ decays with distance $|k-j|$ (approximately exponentially, as shown in Figure 5).

9.2 In the eigenbasis

From Theorem 5.1:

$$\tilde{S}_m(\tilde{\mathbf{x}}, t) = - \frac{\tilde{x}_m - \tilde{\mu}_{t,m}}{e^{-2t}d_m + \Delta_t}. \quad (30)$$

Each mode is an independent scalar denoiser. The “effective SNR” of the m -th mode is:

$$\text{SNR}_m(t) = \frac{e^{-2t}d_m}{\Delta_t}. \quad (31)$$

When $\text{SNR}_m(t) \gg 1$, the m -th mode still carries useful signal and the score is informative. When $\text{SNR}_m(t) \ll 1$, the mode is dominated by noise and the score simply pushes $\tilde{x}_m$ toward the mean.

9.3 Connecting the two bases: the score in frame space from the spectral decomposition

Using (13):

$$S_k(\mathbf{x}, t) = - \sum_{m=0}^K \frac{1}{e^{-2t}d_m + \Delta_t} \sum_{j=0}^K u_m(k) u_m(j) (x_j - \mu_{t,j}). \quad (32)$$

This decomposes the coupling $(\Sigma_t^{-1})_{kj}$ into spectral contributions:

$$(\Sigma_t^{-1})_{kj} = \sum_{m=0}^K \frac{u_m(k) u_m(j)}{e^{-2t}d_m + \Delta_t}. \quad (33)$$

The entry $(\Sigma_t^{-1})_{kj}$ is a weighted sum of products $u_m(k) u_m(j)$ over all modes. For the tridiagonal Markov case, the eigenvectors $u_m(k)$ are spatially localised (sine-like), and nearby frames share similar eigenvector values. This is why $|(\Sigma_t^{-1})_{kj}|$ decays with distance $|k-j|$ even for $t > 0$.

10 Why the Inverse of a Tridiagonal Matrix Is Full

This section explains the algebraic mechanism behind the filling of zeros in Σ_t^{-1} , a point raised by Mézard during the meeting.

Proposition 10.1 (Inverse of a tridiagonal matrix is full). *Let T be an $n \times n$ symmetric, positive definite, tridiagonal matrix with non-zero off-diagonal entries. Then $(T^{-1})_{jk} \neq 0$ for all j, k .*

Proof sketch. Write $T = LDL^\top$ (Cholesky or $LDL^\top$ decomposition). Since T is tridiagonal, L is lower bidiagonal: $L_{ij} \neq 0$ only for $j = i$ or $j = i - 1$. The inverse $T^{-1} = L^{-\top} D^{-1} L^{-1}$. The inverse of a bidiagonal matrix is *full* (lower/upper triangular with all entries non-zero), and the product of two full triangular matrices is full. $\square$

Remark 10.2 (Connection to Fourier modes). *As Mézard noted, one can understand this in Fourier space. The tridiagonal matrix T has eigenvalues $\lambda_m = a + 2b \cos \theta_m$ (approximately), which are all distinct. The inverse has eigenvalues $1/\lambda_m$, but the eigenvectors are shared. Expressing T^{-1} as a sum over eigenvector outer products $\sum_m (1/\lambda_m) \mathbf{u}_m \mathbf{u}_m^\top$, every entry receives a contribution from every mode (since $u_m(k) \neq 0$ for all k, m in general). No cancellation makes $(T^{-1})_{jk}$ exactly zero.*

Now, Σ_t^{-1} involves the inverse of $B = e^{-2t}I + \Delta_t \Sigma_0^{-1}$, which is tridiagonal (a diagonal shift of a tridiagonal matrix is still tridiagonal). By Proposition 10.1, B^{-1} is full, and the product $\Sigma_0^{-1} \cdot B^{-1}$ is full. This is the precise mechanism by which the diffusion noise destroys the tridiagonal sparsity pattern.

11 Summary of Key Results

Main findings

1. Four representations of Σ_t^{-1} :

$$\Sigma_t^{-1} = \frac{1}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1} \quad (\text{Factored}) \quad (34)$$

$$= \Sigma_0^{-1} (e^{-2t} I + \Delta_t \Sigma_0^{-1})^{-1} \quad (\text{Mézard}) \quad (35)$$

$$= \frac{1}{\Delta_t} I - \frac{e^{-2t}}{\Delta_t^2} \Sigma_0 (\Sigma_0^{-1} + \frac{e^{-2t}}{\Delta_t} I)^{-1} \quad (\text{Woodbury}) \quad (36)$$

$$= U \text{diag}(1/(e^{-2t} d_m + \Delta_t)) U^\top \quad (\text{Spectral}) \quad (37)$$

2. At $t = 0$: Markov structure. Σ_0^{-1} is tridiagonal—nearest-neighbour coupling only.

3. At $t > 0$: zeros fill in. The inversion in Mézard’s form (35) maps the tridiagonal structure to a full matrix. The beyond-tridiagonal entries decay approximately exponentially with frame distance and with e^{-2t} .

4. Tridiagonal weight fraction $\rho_{\text{tri}}(t)$ decays monotonically from 1 (at $t = 0$) toward $1/(K + 1)$ (at $t \rightarrow \infty$, uniform over all entries of I). The Markov “fingerprint” persists most strongly at small t and for large α .

5. Shared eigenvectors. Σ_t , Σ_t^{-1} , and Σ_0 all share the same eigenvectors U , for all t . Only the eigenvalues change: $\lambda_m(t) = e^{-2t} d_m + \Delta_t$.

6. Full decoupling in the eigenbasis. The rotated score $\tilde{S}_m = -(\tilde{x}_m - \tilde{\mu}_{t,m})/(e^{-2t} d_m + \Delta_t)$ is independent across modes. The reverse SDE splits into $K + 1$ independent scalar processes.

7. This decoupling is Gaussian, not Markov. Any Gaussian P_0 (with any covariance structure) admits an eigenbasis in which Σ_t^{-1} is diagonal. The Markov property determines the specific form of the eigenvectors (sine-like, spatially localised) and eigenvalues (the formula (15)), but does not by itself enable the decoupling.

12 Discussion and Next Steps

12.1 What Markovianity contributes (and what it does not)

The analysis reveals a precise decomposition of roles:

Property	Source
Σ_0^{-1} is tridiagonal	Markov property of $P_0(\mathbf{a})$
Σ_t and Σ_t^{-1} share eigenvectors with Σ_0	Gaussianity of $P_0(\mathbf{a})$ + i.i.d. OU noise
Score decouples in eigenbasis	Gaussianity of $P_0(\mathbf{a})$
Eigenvectors are sine-like / spatially localised	Tridiagonal (Markov) structure of Σ_0^{-1}
Eigenvalue spectrum has specific form (15)	AR(1) dynamics (value of α , σ_η^2)
Off-diagonal entries of Σ_t^{-1} decay with distance	Eigenvector localisation (indirect Markov effect)

The key takeaway: Markovianity provides the *specific structural form* of the eigenvectors and eigenvalues, but the *existence* of a fully decoupled basis is a generic Gaussian phenomenon.

12.2 Next steps

Following Mézard’s roadmap from the meeting:

- (i) **Enrich $P_0(\mathbf{a})$** : give P_0 its own non-trivial covariance structure within each frame (multivariate frames), to disentangle the effect of intra-frame correlations from the Markov inter-frame structure.
- (ii) **Mixture of Gaussians**: replace the Gaussian P_0 with a mixture. The score becomes nonlinear in $\mathbf{x}$, and the eigenbasis decoupling breaks. The question is: does the Markov structure of the score (nearest-neighbour dominance) survive in the non-Gaussian case?
- (iii) **Quantify the “Markov residual” in the score**: define indicators that measure how close Σ_t^{-1} is to tridiagonal as a function of t , beyond the simple weight fraction $\rho_{\text{tri}}(t)$ used here.
- (iv) **Connect to the Kalman smoother**: the backward pass of the Kalman smoother is the propagator that computes the posterior mean $\langle a_k \rangle_{\mathbf{a}|\mathbf{x}}$ in $O(K)$ time. Understand how the smoother gain relates to the spectral decomposition.

13 Notation Reference

Symbol	Meaning
K	Number of AR(1) steps; $K + 1$ frames total (indices $0, \dots, K$)
$t \geq 0$	Diffusion (OU noise) time
$\alpha \in (-1, 1)$	AR(1) contraction coefficient
σ_0^2	Variance of initial frame a_0
σ_η^2	AR(1) innovation variance
$\sigma_\infty^2 = \sigma_\eta^2 / (1 - \alpha^2)$	Stationary variance
$\Delta_t = 1 - e^{-2t}$	OU noise variance at diffusion time t
$\gamma_t = e^{-2t} / \Delta_t$	Signal-to-noise ratio
Σ_0	$(K+1) \times (K+1)$ clean covariance; $(\Sigma_0)_{jk} = \sigma_\infty^2 \alpha^{ j-k }$ (stationary)
$\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I$	Noisy covariance
$\boldsymbol{\mu}_a, \boldsymbol{\mu}_t = e^{-t} \boldsymbol{\mu}_a$	Clean / noisy mean
d_m	m -th eigenvalue of Σ_0 ($d_0 \geq d_1 \geq \dots \geq d_K$)
$\mathbf{u}_m$	m -th eigenvector of Σ_0 (shared by Σ_t and Σ_t^{-1} for all t)
$U = [\mathbf{u}_0 \dots \mathbf{u}_K]$	Orthogonal eigenvector matrix
$\lambda_m(t) = e^{-2t} d_m + \Delta_t$	m -th eigenvalue of Σ_t
$\tilde{\mathbf{x}} = U^\top \mathbf{x}$	Noisy trajectory in the eigenbasis
$S(\mathbf{x}, t) = -\Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t)$	Joint score
$\tilde{S}_m = -(\tilde{x}_m - \tilde{\mu}_{t,m}) / \lambda_m(t)$	Score in the eigenbasis (independent per mode)
$\rho_{\text{tri}}(t)$	Tridiagonal weight fraction of Σ_t^{-1}